PairwiseDistance#

https://pytorch.org/docs/stable/generated/torch.nn.modules.distance.Pairwise

Description:

Computes the pairwise distance between input vectors, or between columns of input matrices. Distances are computed using p-norm, with constant eps added to avoid division by zero if p is negative, i.e.: where eee is the vector of ones and the p-norm is given by. p (real, optional) – the norm degree. Can be negative. Default: 2 eps (float, optional) – Small value to avoid division by zero. Default: 1e-6

Function Signature

```
class torch.nn.modules.distance.PairwiseDistance(p=2.0,
eps=1e-06, keepdim=False)[source]#

Parameters

Shape:

forward(x1, x2)[source]#

Return type
```

Returns:

Tensor

Code Examples

```
>>> pdist = nn.PairwiseDistance(p=2)
>>> input1 = torch.randn(100, 128)
>>> input2 = torch.randn(100, 128)
>>> output = pdist(input1, input2)
```

Detailed Documentation

Documentation

Computes the pairwise distance between input vectors, or between columns of input matrices.

Distances are computed using p-norm, with constant eps added to avoid division by zero if p is negative, i.e.:

where eee is the vector of ones and the p-norm is given by.

p (real, optional) – the norm degree. Can be negative. Default: 2

eps (float, optional) – Small value to avoid division by zero. Default: 1e-6

keepdim (bool, optional) – Determines whether or not to keep the vector dimension. Default: False

Input1: (N,D)(N, D)(N,D) or (D)(D)(D) where N = batch dimension and D = vector dimension

Input2: (N,D)(N, D)(N,D) or (D)(D)(D), same shape as the Input1

Output: (N)(N)(N) or ()()() based on input dimension. If keepdim is True, then (N,1)(N, 1)(N,1) or (1)(1)(1) based on input dimension.

Runs the forward pass.